Raúl de la Fuente

Email | Website | GitHub | LinkedIn

Urbana, IL

Summary

I am a first-year Computer Science Ph.D. student at the University of Illinois Urbana-Champaign. My research interests lie at the intersection of microarchitecture and 3D VLSI integration. I am currently exploring System-Technology Co-Optimization (STCO) methodologies to design next-generation heterogeneous systems—including CPUs, GPUs, and accelerators—that leverage vertical scaling to minimize latency, maximize interconnect density, and optimize energy efficiency.

Education

University of Illinois Urbana-Champaign *Fall 2025 - Present*
Ph.D. | *Computer Science* *Urbana, IL*
Advisor: Prof. Josep Torrellas

University of Chile *Fall 2023 - Spring 2024*
M.S. | *Computer Science* *Santiago, Chile*
Advisors: Prof. Alejandro Hevia and Prof. Luciano Radrigan

University of Chile *Fall 2018 - Fall 2022*
B.Eng. | *Computer Science and Engineering* *Santiago, Chile*

Relevant Coursework

Spring 2026: CS 533 - Parallel Computer Architecture, ECE 425 - VLSI Design
Fall 2025: CS 433 - Computer System Organization, CS 426 - Compiler Construction, CS 484 - Parallel Programming

Publications

Enhancing Predictive Maintenance in Mining Mobile Machinery through a TinyML-enabled Hierarchical Inference Network. *Raúl de la Fuente,* Luciano Radrigan and Anibal S Morales. In *IEEE Access* (**IEEE Access**).

Poster abstract: Hands-on Evaluation of Kinéis Satellite IoT Technology. *Raúl de la Fuente and* Thomas Watteyne. In *Proceedings of the 22nd International Conference on Information Processing in Sensor Networks* (**IPSN '23**).

Research Experience

I-acoma Group, UIUC *Urbana, IL*
Graduate Research Assistant | *Advisor: Prof. Josep Torrellas* *August 2025 - Present*

- Conducting literature review on state-of-the-art 3D VLSI integration and STCO, focusing on CMOS 2.0-enabled microarchitectural partitioning strategies for logic-on-logic stacking.

AIO Team, INRIA *Paris, France*
Research Intern | *Advisor: Prof. Thomas Watteyne* *January 2023 - April 2023*

- Deployed a low-power mesh IoT network based on Dust Networks' SmartMesh IP technology for environmental monitoring in urban areas.

- Conducted a hands-on evaluation of a state-of-the-art satellite communication module for IoT applications, assessing its performance and energy efficiency.

Dept. of Electrical Engineering, University of Concepción *Concepción, Chile*
Research Assistant | Advisor: Prof. Luciano Radrigan *July 2022 - December 2024*

- Deployed a cyber-physical system for real-time monitoring of industrial processes using on-device machine learning for anomaly detection and failure prediction.
- Mainly responsible for: (i) Implementing the nodes' firmware in C/C++ , (ii) Designing and optimizing neural networks for embedded devices, and (iii) Conceptualizing an algorithm to dynamically switch between cloud and edge inference based on network conditions.

CLCERT Group, University of Chile *Santiago, Chile*
Research Assistant | Advisor: Prof. Alejandro Hevia *January 2022 - December 2024*

- Contributed to the Psifos voting system, leveraging homomorphic encryption and zero-knowledge proofs to ensure secure and verifiable electronic elections.
- Migrated the system from traditional finite-field cryptography to elliptic curve cryptography, enhancing security and computational efficiency.
- Optimized the distributed key generation protocol used by election trustees to reduce computational overhead and enable participation from resource-constrained embedded devices.

NIC Chile Research Labs *Santiago, Chile*
Research Intern | Mentor: Eduardo Riveros *January 2021 - March 2021*

- Pentested the internal networks of NIC Chile and the Faculty of Physical and Mathematical Sciences at the University of Chile, identifying and reporting critical vulnerabilities.
- Developed a plugin for the Nmap network scanner to identify vulnerable IoT devices within nested networks, enabling automated scanning and reporting for security audits.

Activities

November'24	Presented talk 'Enhancing Predictive Maintenance in Mining Mobile Machinery through a TinyML-enabled Hierarchical Inference Network' at the CS Master's Seminar, University of Chile.
2023 - 2024	Served as Teaching Assistant for both the Internet of Things Systems Design and Web Application Development courses at University of Chile.
May'23	Presented the poster 'Hands-on Evaluation of Kinéis Satellite IoT Technology' at IPSN '23, San Antonio, TX.
January'23	Assisted to the OpenSwarm project launch event at INRIA, Paris, France.
2022	Participated in over 250 elections powered by the Psifos voting system, working with both internal departments of the University of Chile and external organizations.
2021	Served as Teaching Assistant for the Systems Programming course at University of Chile.

Awards

- Recipient of the Master's in CS Scholarship from the University of Chile for the academic years 2023-2024.
- Recipient of the Academic Excellence Award from the University of Chile for the years 2018-2022.

Technical Skills

Hardware: SystemVerilog | Vivado | Gem5 | Ramulator | McPAT | ESP32 | nRF52 | Raspberry Pi
Programming: C/C++ | Python | Rust | RISC-V | GoLang | Java | Scala
Software: Docker | Bash | Git | TensorFlow | Pytorch | LLVM IR | MPI | OpenMP | Valgrind